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Investigating Machine Learning Driven Metrics to Evaluate Research Quality

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Abstract

The rapid growth of scientific literature has made it increasingly difficult to assess the quality and impact of academic works using metrics like the H-index. In this work, we explore the use of Large Language Models and text embeddings to investigate new metrics to evaluate the impact of academic authors and their publications. Paper abstracts were retrieved from the OpenAlex API and turned into vectors using `all-mpnet-base-v2` and `E5-Base-V2`, then compared via pairwise cosine similarity. We develop two pipelines: one applied to 1195 McGill University Physics researchers, and one applied to a dataset of 9417 Nuclear and High Energy Physics papers. For the author-level analysis, we find that the mean and distribution of Self-Similarity scores vary significantly between researchers with comparable H-indices, suggesting that embedding-based similarity captures a dimension of academic output that conventional metrics overlook. A weak negative correlation between average Self-Similarity and H-index is observed across the authors. The paper-level analysis shows potential for a data screening tool to identify outliers within a particular field. These results suggest that AI-driven semantic analysis may enable more comprehensive assessments of scientific impact.

1 Introduction

The exponential growth of scientific literature is increasing the workload involved in assessing the academic impact of authors and their works. The number of peer-reviewed articles has been growing faster than the number of qualified peer reviewers, partly driven by the 'publish or perish' mindset in which researchers are incentivized to maximize publication counts to improve perceived productivity and funding prospects [1]. One of the most widely used metrics for research output quality is the Hirsch index (H-index): the largest n such that an author has at least n papers each cited at least n times [2]. The H-index favors researchers in high-volume fields, disadvantages early-career scientists, and fails to capture the conceptual depth or originality of a body of work.

This project explores the application of machine learning techniques to generate metrics that allow for more in-depth assessments of research impact, specifically through natural language processing (NLP) using text embeddings [3, 4] to calculate the cosine similarity between a given pair of texts. We calculated cosine similarity between a given author's publications (Self-Similarity) and between papers within a particular field (comparison by field). The process of embedding a text is that the model clusters them within a vector space to capture the meaning of the text, which is determined by the unique training process of each model, and the cosine allows us to calculate how similar a pair of vectorized texts are. The cosine similarity between embedding vectors A and B is:

$$\cos\theta = \frac{A \cdot B}{\|A\| \cdot \|B\|}. \tag{1}$$

Accurate similarity scores depend on the embedding method, which varies across models. A model that captures only surface lexical similarity will assign high scores to papers that share terminology but differ in meaning, and low scores to papers on the same topic expressed in different words [5]. For the prior mentioned metrics to be meaningful, the embeddings must respond to *conceptual* rather than merely *terminological* overlap [6, 7]. An ideal model should satisfy identical > synonyms > antonyms/homonyms > unrelated. Prior work has explored AI-estimated article quality for ranking scholarly documents [8], suggesting that with the right model, machine learning approaches hold potential to complement conventional metrics.

2 Materials and Methods

The dataset used for the Self-Similarity portion was comprised of 1195 McGill-affiliated researchers in the field of Physics and Astronomy, while the dataset used for the comparison by field portion contained 9417 papers in Nuclear and High Energy Physics (NHE dataset). We also employed a small dataset of 1602 Astronomy and Astrophysics (AA) papers to compare the different models. Due to the limited availability of complete full papers, analyses were restricted to the abstracts of the selected papers.

2.1 Data collection via OpenAlex

All datasets were obtained from OpenAlex [9], an open-access bibliographic database. The fields extracted for each author were: author ID, fields of study, institution affiliations, publication count, citation count, H-index, and a complete list of all their published works available through OpenAlex. The fields extracted for each work were: OpenAlex ID, DOI, title, abstract, author information, publication year, and citation count. The data was acquired using a collection script [10] that was modified to suit the needs of the project.

2.2 Model Comparison

To determine which models to use, we selected four candidate models and compared their similarity scores for the 11 pairs of sentences in Table 1. The sentence pairs are categorized into identical, synonymous, antonymous, homonymous, and unrelated pairs.

Table 1: Benchmark Sentences for Text Similarity Evaluation.

ID	Sentence A	Sentence B	Relationship
0	The quick brown fox jumps.	The quick brown fox jumps.	Identical
1	A person is slicing an onion.	Someone is cutting an onion.	Synonyms
2	The movie was incredible.	The film was great.	Synonyms
3	I forgot my password.	How do I recover my password?	Synonyms
4	He is a professional athlete.	He plays sports for a living.	Synonyms
5	I love eating green apples.	I enjoy eating red apples.	Synonyms
6	The cat ate the mouse.	The mouse ate the cat.	Antonym
7	Python is a programming language.	A python is a large snake.	Homonym
8	The bank is closed today.	I sat on the river bank.	Homonym
9	The sky is blue today.	I need to buy new shoes.	Unrelated
10	Stock prices are rising.	Please pass the salt.	Unrelated

The four models evaluated were: (1) **SPECTER** [11], pre-trained on scientific citation graphs; (2) **E5-Base-V2** [12], a general-purpose bi-encoder from Microsoft; (3) **Embedding Gemma**, accessed via Ollama [13]; and (4) **all-mpnet-base-v2** [14], producing 768-dimensional embeddings, ranked among the top ten on the MTEB benchmark (found in Appendix A). The results were normalized to allow for direct comparison since every model had a different range of values. The normalization method was to force all plots to start at $y = 1$ and end at $y = 0$, and rescale the remaining y values using the formula:

$$y_{normalized} = \frac{y - y_{end}}{y_{start} - y_{end}} \tag{2}$$

where y_{start} and y_{end} are the maximum and minimum values, respectively, for a given plot.

We then tested the different models on the set of 1602 Astronomy and Astrophysics papers to gauge performance in a setting closer to our area of focus. We implemented the K-nearest neighbor (KNN) algorithm [15] to find the 5 most similar neighbors for each paper. We then calculated the mean of the 5 scores for all 4 models, normalized the results, and plotted them as seen in Figure 2.

2.3 Comparison of Papers by Field

We ran the KNN algorithm on the NHE dataset using E5-Base-V2 with a configurable author-overlap ratio threshold, where the author overlap is defined as:

$$\text{Overlap}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)} \tag{3}$$

where A and B are the sets of authors of the two papers being compared. Rather than excluding any paper sharing a single author, the ratio threshold allows partial overlap to accommodate large collaborative papers. A strict exclusion criterion would unnecessarily discard papers that are genuinely topically related but happen to share one or two authors out of a large team. By varying the threshold τ, we allow flexibility across different authorship conventions and collaboration scales. Figure 3 shows the effect of varying τ on the average neighbor similarity.

2.4 Author Self-Similarity

For each author that met the aforementioned conditions, the Self-Similarity was computed as the average similarity of each unique abstract pair within the author's list of publications. The Self-Similarity and standard deviation σ_s for each author were recorded in a CSV file alongside all the other data collected from OpenAlex, all of which can be found on our

GitHub [16]. To ensure the quality of the data collected from OpenAlex, uniqueness was determined by tracking the titles and DOIs of the works. Abstracts containing MathML-encoded equations were filtered out before embedding, as current models cannot meaningfully process this format. Only authors with at least 3 unique abstracts without MathML-encoded equations were included in the analysis dataset.

3 Results

3.1 Model Comparison

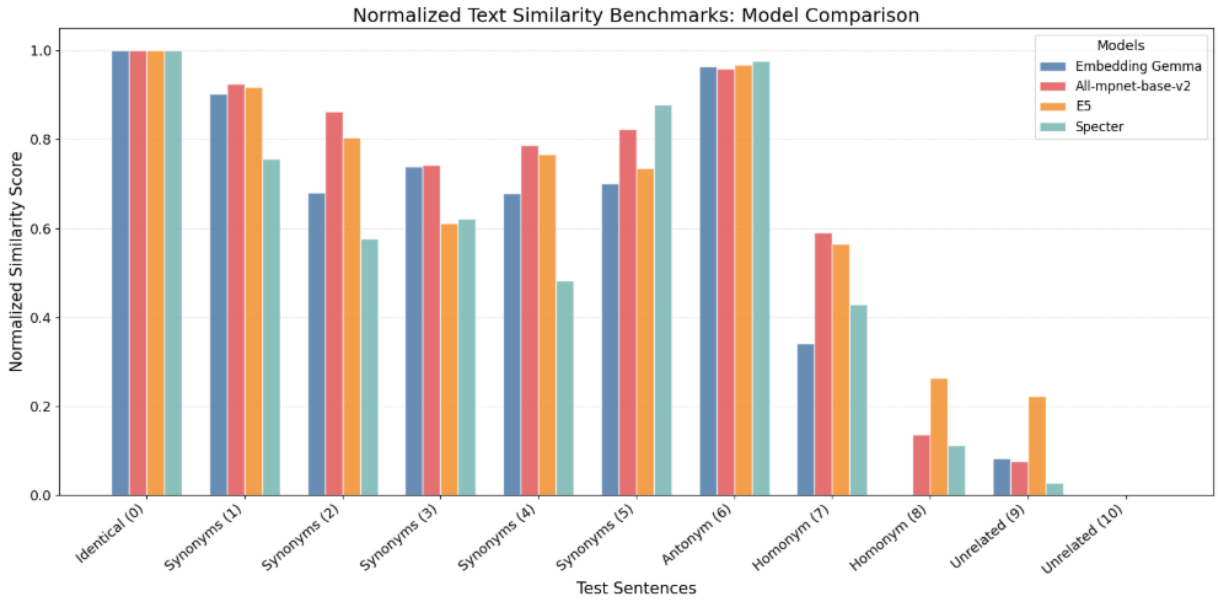


Figure 1: Normalized cosine similarity scores for the four candidate embedding models. The models (SPECTER, E5-Base-V2, `all-mpnet-base-v2`, and Embedding Gemma) were tested on the eleven benchmark sentence pairs of Table 1. Scores are normalized per model using Equation (2) so that the maximum and minimum values map to 1 and 0, respectively, enabling direct comparison across models. The unnormalized plot can be found in Appendix A.

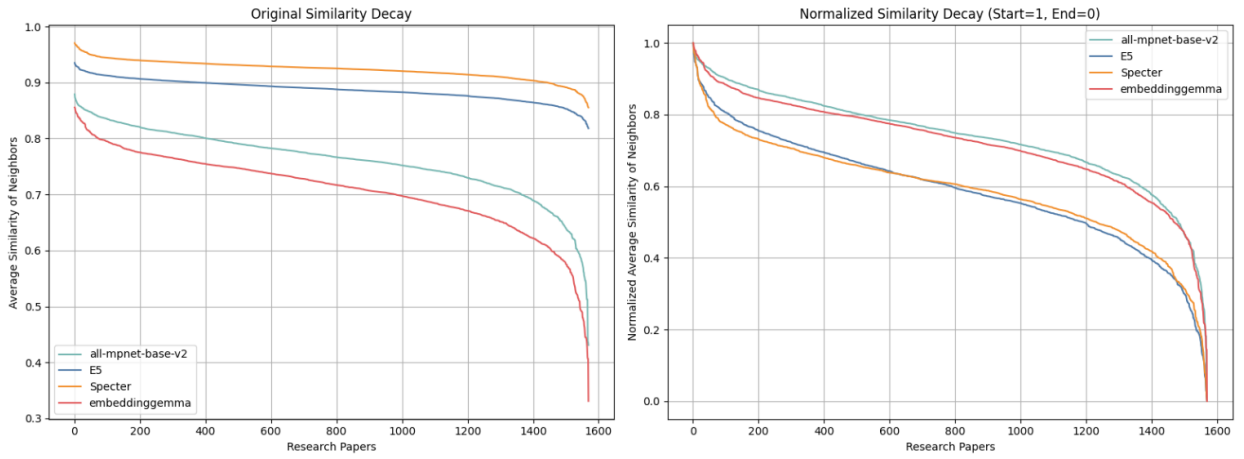


Figure 2: Unnormalized (left) and normalized (right) average similarity to the 5-nearest neighbor for each of the four embedding models on the AA dataset. The left panel shows absolute score ranges; the right panel normalizes the data using Equation (2) to allow for direct comparison.

The average similarity decays as neighbors become more distant for the 4 models (Figure 2, left). Each model has their own range of values. Models like `Specter` and `E5` sit higher on the y-axis and span a very small range of values compared to `all-mpnet-base-v2` and `embeddinggemma`. However, this partly reflects the model's scoring scale rather than embedding quality.

We used the same normalization equation (2) to allow for a fair comparison of the decay shape (Figure 2, right). The difference between the plots does not dictate the relative performance of our selected models. They just outline how the abstracts are distributed in the embedding space.

For the analysis of our two datasets, we chose to use `E5-Base-V2` for the NHE dataset and `all-mpnet-base-v2` for the Self-Similarity analyses. We picked `E5-Base-V2` because it is the highest rated in Table 2.2 among the 4 tested, and we chose to use `all-mpnet-base-v2` due to computational resource limits of the computer used.

3.2 Comparison of Papers by Field

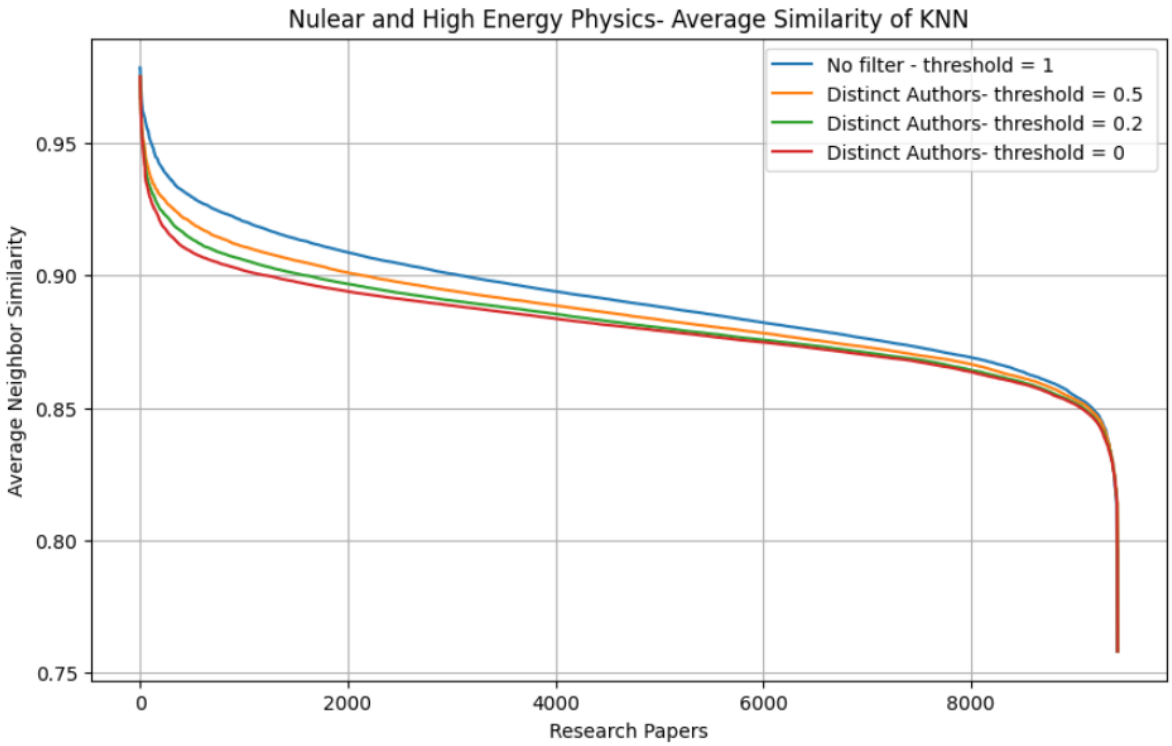


Figure 3: Average similarity of the 5 nearest neighbors for papers in the Nuclear and High Energy Physics (NHE) dataset, computed with E5-Base-V2 embeddings. The thresholds τ are based on Equation (3). At $\tau = 0$, any paper sharing even a single author with the query is excluded; at $\tau = 1$, no filtering is applied.

The decrease in average similarity as τ decreases suggests that papers by overlapping author groups are more topically similar than the broader field average. The rate of decline with τ quantifies how much of the unfiltered KNN similarity is attributable to coherence within the group rather than genuine topical proximity across the field. As the number of authors in $min(|A|, |B|)$ increases, τ can be increased accordingly to avoid unnecessary exclusions.

3.3 Author Self-Similarity

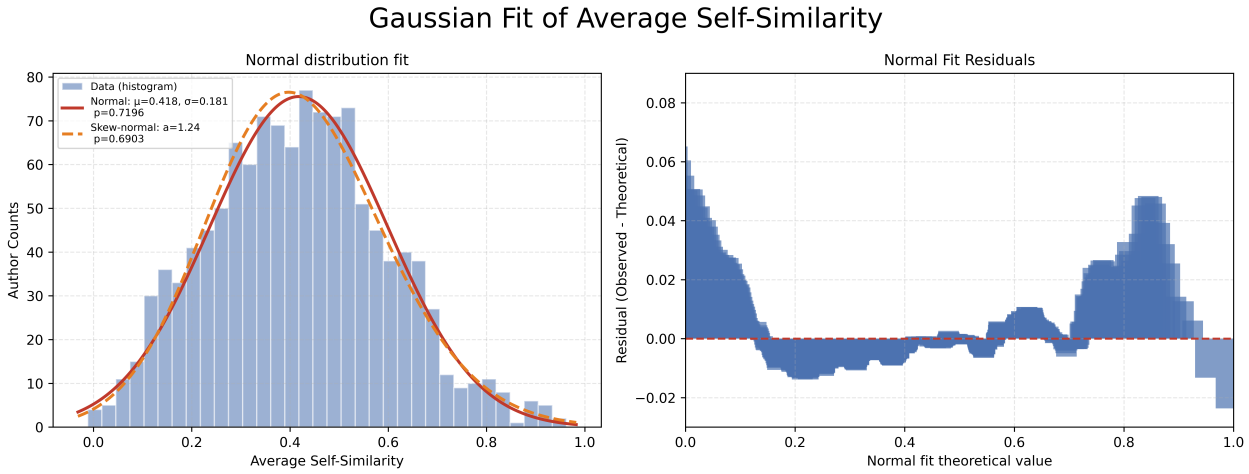


Figure 4: Author Self-Similarity is normally distributed. **Left:** The distribution of author Self-Similarity (blue) with a fitted Normal distribution ($\mu = 0.418$, $\sigma = 0.181$; red) and a Skew-Normal distribution ($a = 1.24$, orange dashed). For each fit, the p-value is shown in the legend, showing how closely the data fits the distribution. **Right:** residuals between the sorted observed similarity values and the theoretical Normal quantiles.

The distribution of Self-Similarity within the sample of 1195 authors is well described by both the normal and skew-normal distributions ($p > 0.50$). The best fit normal distribution has an average Self-Similarity value of 0.418 ± 0.181 ($p = 0.7196$), while the skew-normal distribution has an average Self-Similarity of $x \pm y$ ($p = z$).

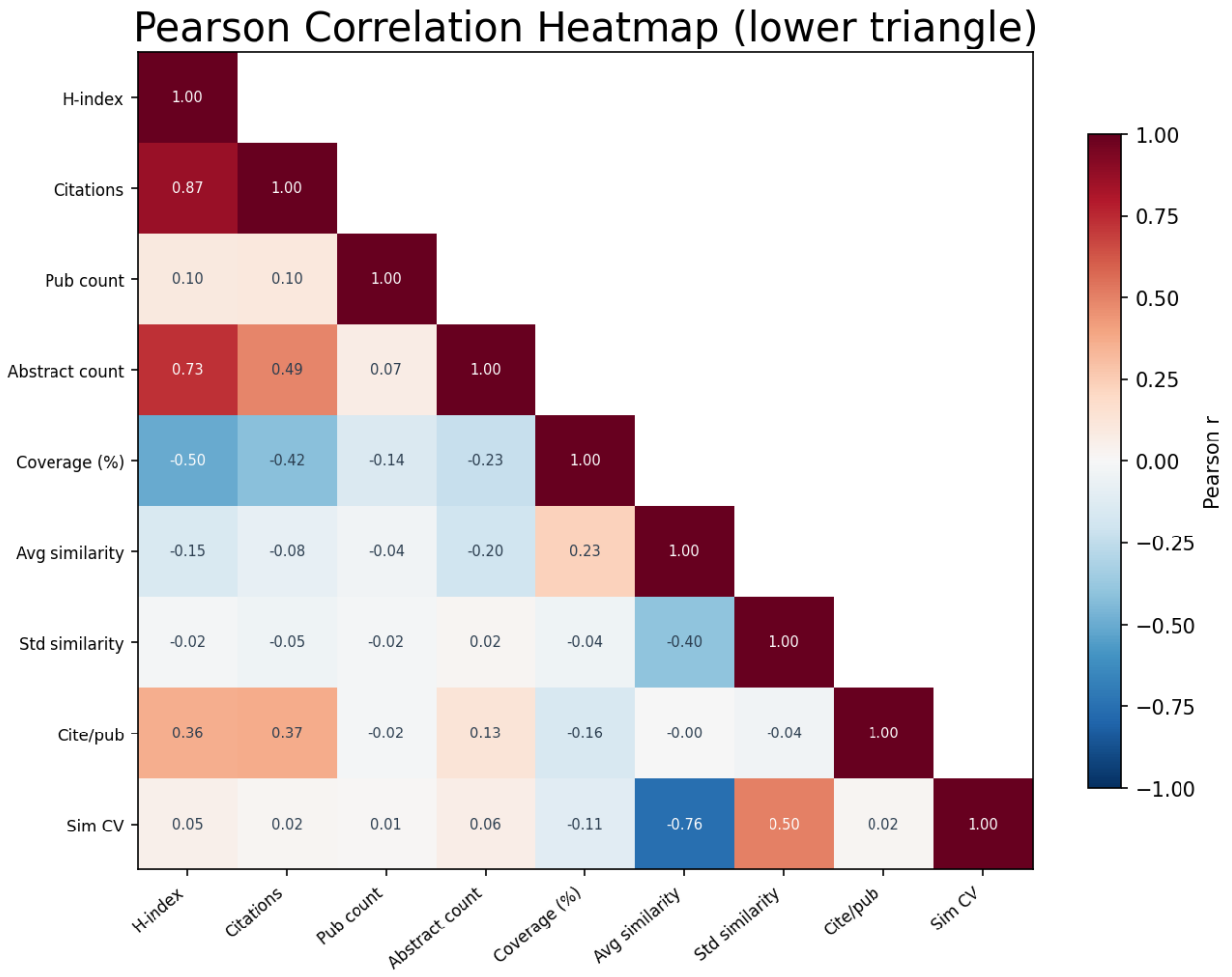


Figure 5: Pearson correlation heatmap for nine bibliometric variables computed across the Self-Similarity dataset (lower triangle only). Red = positive, blue = negative, white ≈ zero, on a scale of −1 to 1.

Average Self-Similarity has a correlation value of 0.23 with coverage, a correlation value of −0.20 with abstract count, a correlation value of −0.15 with H-index, and a correlation value of −0.40 with its standard deviation. The remaining correlation values are either close to 0 or expected to be perfectly positively correlated (e.g. H-index to H-index) or tautologically correlated (e.g. H-index to citation count).

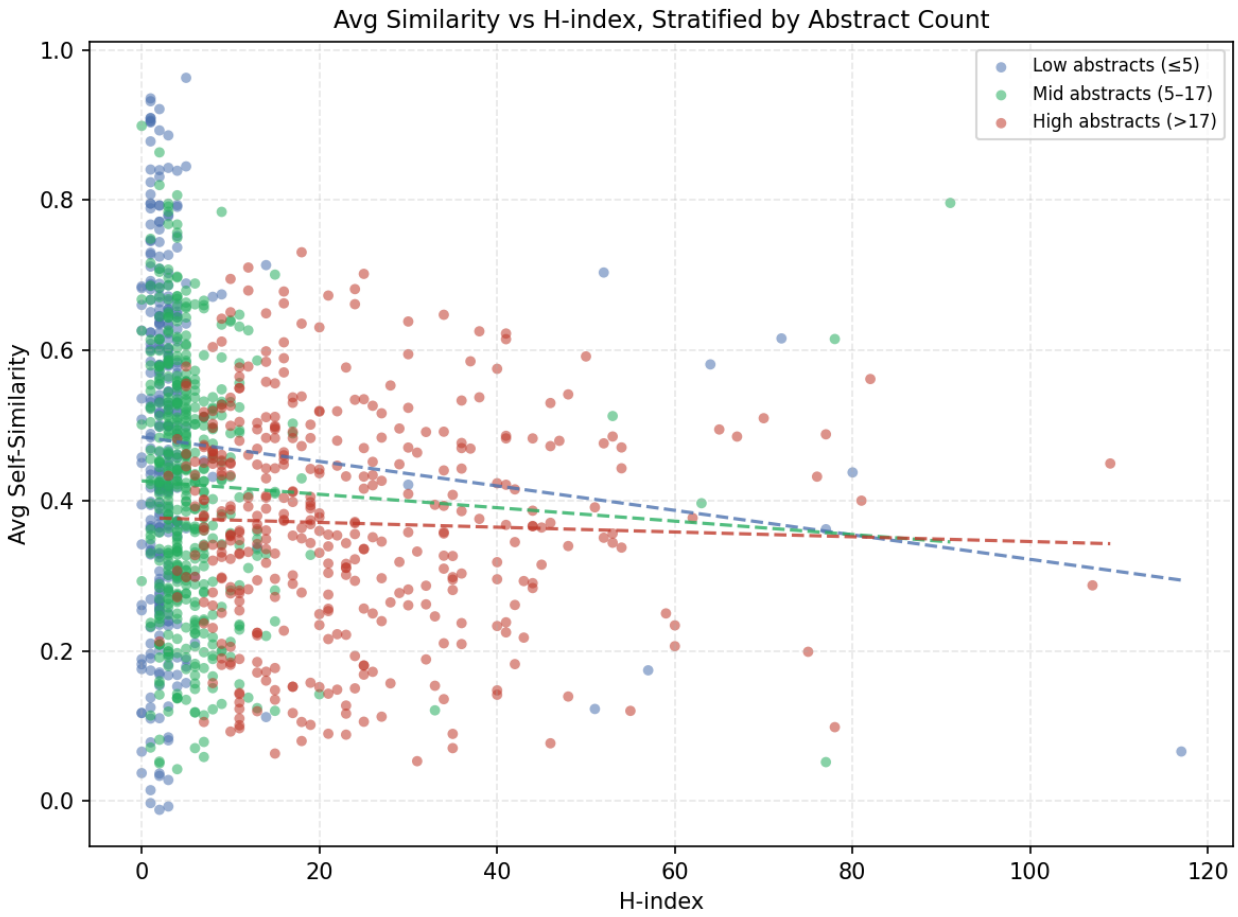


Figure 6: Correlation between average Self-Similarity and H-index decreases with increasing available abstract count. Average Self-Similarity plotted against H-index, with authors evenly divided into three equal groupings of abstract count ranges: $3 \leq blue \leq 5 \leq green \leq 17 \leq red$. Each group is accompanied by an independent linear trend line.

Dividing the dataset into three equal sets based on abstract count, the number of abstracts available through OpenAlex, allows us to visualize the effect that greater access to abstracts would have on the calculated correlations. For authors with more abstracts available, the correlation between average Self-Similarity and H-index weakens, suggesting that data availability in OpenAlex introduces a confounding bias (Figure 6).

4 Discussion

This work investigated whether machine learning–based text embeddings can produce meaningful metrics for evaluating academic research beyond traditional bibliometrics. Using abstracts retrieved from OpenAlex for 1195 McGill Physics researchers and 9417 Nuclear and High Energy Physics papers, we applied two sentence-transformer models, `all-mpnet-base-v2`

and `E5-Base-V2` respectively, to compute pairwise cosine similarity scores. Using these cosine similarity scores, this project analyzed the field-level KNN similarity decay of the Nuclear and High Energy Physics papers, and the author Self-Similarity of the 1195 McGill Physics researchers.

4.1 Comparison by Field

A potential application of the approach in Figure 3 is to flag papers at both extremities of the similarity decay curve for further investigation. At the left extremity, papers with exceptionally high neighbor similarity despite distinct authorship, particularly when $\tau = 0$, could indicate plagiarism or warrant investigation into predatory publishing practices. Only abstracts were compared, however, so high similarity does not necessarily indicate misconduct: papers with highly similar abstracts could contain distinct contributions in the body. At the right extremity, papers with unusually low neighbor similarity may represent interdisciplinary works sitting at the boundary of the target field and other fields, innovative papers, or low-quality and incomplete abstracts. This was exemplified during our analysis, with some entries not corresponding to abstracts (parsing error) or belonging to different fields (misclassification error), likely the result of extraction errors from OpenAlex, where metadata is occasionally misattributed or incomplete. These findings suggest that the similarity decay curve could serve as a lightweight data quality screening tool, complementing more manual curation approaches.

4.2 Author Self-Similarity

The distribution of Self-Similarity, shown in Figure 4, follows a normal probability distribution, implying that it would make a well-balanced metric that can capture outliers in both the high Self-Similarity and low Self-Similarity ranges.

In Figure 5, the strong positive correlations between H-index, average citations per publication, citation count, and abstract count are expected and serve as a data-quality check, since it is expected that researchers with a higher H-index are cited more on each of their publications, and are more likely to have their works archived by sites like OpenAlex. The Pearson correlation heatmap also reveals a weak negative association between average Self-Similarity and conventional metrics like H-index, citation count, and publication count, indicating that an author's Self-Similarity is not well reflected in these conventional metrics. The correlation between Self-Similarity and coverage or abstract count, 0.23 and -0.20 respectively, suggests potential sources of bias introduced by the data acquisition strategy used by OpenAlex.

Similarly, Figure 6 shows that authors with more abstracts available have less significant correlation between Self-Similarity and H-index. This supports the interpretation that with

a more complete data set, a full body of works for each author, the observed correlations between Self-Similarity and conventional metrics will be further diminished.

Altogether, these findings suggest that Self-Similarity is largely independent of conventional metrics used to quantify researchers' contributions to academia, indicating that it may offer complementary insight into research impact. High Self-Similarity could indicate deep expertise within a focused subfield or repetitive publication to inflate output counts. Conversely, low Self-Similarity may reflect a meaningful interdisciplinary scope of studies, an early-career researcher who is trying to figure out what they want to study, or a senior scientist leading a lab with a broad scope. Without additional context, Self-Similarity cannot meaningfully distinguish between these possible interpretations. As such, it is best applied as a complementary metric that accentuates, rather than replaces, the insight granted by conventional metrics.

4.3 Limitations

Model limitations: A notable limitation of the models is their tendency to assign higher similarity scores to antonym pairs than to synonym pairs, as seen in Antonym(6) in Figure 1. This suggests that the models rate sentences based on a blend of lexical and semantic similarity rather than purely semantic content. While this behaviour is evident in the benchmark results, it is unlikely to meaningfully impact performance on the target dataset, as scientific abstracts rarely contain sentences that are lexically similar but semantically opposed. This limitation is partly attributable to the models' relatively modest performance rankings (Table 7), which were a consequence of the computational constraints of this project. Given greater resources, a natural improvement would be to employ a more powerful model, or one pretrained on scientific literature such as **SPECTER2** [17] — the successor to SPECTER — to produce embeddings grounded in purely semantic similarity. For broader real-world applications, fine-tuning the models on domain-specific data would further improve their ability to capture semantic relationships.

Data acquisition: Data acquired through OpenAlex does not necessarily contain all available works for any given author, and even when a paper is cataloged in OpenAlex, there is no guarantee that the abstract is provided or unique. Figure 6 aims to acknowledge this limitation by showing how subsets of the data categorized by abstract count have differing correlations between Self-Similarity and H-index. Nonetheless, it would be necessary to acquire more complete datasets to thoroughly evaluate the validity of these measures.

A further limitation concerns MathML-formatted equations, which the embedding models

used in this project could not interpret meaningfully due to the models' current capabilities, found in several abstracts within our raw dataset. These abstracts were entirely excluded from the datasets to preserve the integrity of the text vectorization, which meant that some scientific data were unavailable for analysis. To address this issue, it would be necessary to convert MathML expressions into a format that the models could meaningfully interpret prior to embedding.

Generalizability: Given that the authorship analysis was restricted to McGill Physics faculty and the paper-level analysis drew exclusively from Nuclear and High Energy Physics papers, the generalizability of these findings is limited. The next steps would be to extend the analysis to multiple institutions and fields, embed full paper texts, and incorporate domain-specific models such as SPECTER2 [17].

5 Conclusion

A key insight from this work is the inherent ambiguity of Self-Similarity as a standalone metric: high values may reflect deep domain expertise or redundant publication strategies, while low values may indicate meaningful interdisciplinary breadth or an unfocused early-career trajectory. This ambiguity reinforces that Self-Similarity is best deployed as a complement to existing metrics rather than a replacement, offering additional insight to interpret patterns that H-index and citation counts leave opaque.

Future work should address the limitations identified here by incorporating full-text embeddings, extending the analysis beyond McGill Physics to multiple institutions and disciplines, and adopting domain-specific models such as SPECTER2 [17]. Doing so would test the generalizability of these results and bring the pipeline closer to a practical research evaluation tool.

6 Contribution of Authors

Benjamin Collins and Mo'tasim Mauthoor contributed equally to this paper. Benjamin led the self-similarity portion of the project, and Mo'tasim managed the paper-level analysis. All writing was produced through equal collaboration.

7 Declaration of AI Use

Generative AI was used to help design the code used in this project. All scientific analysis, interpretation, and writing are the work of the authors

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A Appendix A

Rank	Model	Performance	Strength	Resources Needed	Runs on my PC
1	Voyage-3-large	Excellent (SOTA)	Best overall semantic + retrieval	API (cloud)	No (API only)
2	Gemini-005	Excellent (SOTA)	Strong reasoning-aware embeddings	API (cloud)	No (API only)
3	BGE-M3	Excellent	Multi-lingual + multi-function (dense/sparse)	~8–12GB RAM	Borderline (slow CPU)
4	E5-large-v2	Excellent	Very strong semantic similarity	~6–8GB RAM	Yes (slow)
5	BGE-large-en-v1.5	Excellent	Top-tier retrieval	~6–8GB RAM	Yes (slow)
6	Instructor-XL	Very high	Task-aware embeddings	~10GB+ RAM	Not practical
7	Jina-v3	Very high	Strong long-context + modern design	~4–6GB RAM	Yes
8	Qwen3-8B	High	LLM-based semantic depth	8B params (quantized needed)	Hard but possible
9	E5-base-v2	Very high	Best local balance	~2GB RAM	Yes
10	BGE-base-en-v1.5	Very high	Strong baseline	~2GB RAM	Yes
11	GTR-T5-base / large	High	Good semantic reasoning	4–10GB RAM	Base: Yes, Large: slow
12	SPECTER	High (domain-specific)	Best for citation-style similarity	~2–3GB RAM	Yes
13	Embedding Gemma	Moderate–High	LLM-style embeddings	~4–8GB RAM	Depends
14	all-mpnet-base-v2	Moderate–High	Strong classic baseline	~2GB RAM	Yes
15	all-MiniLM-L6-v2	Moderate	Fast, lightweight	~1GB RAM	Yes

Figure 7: List of top 15 embedding models in descending order of performance

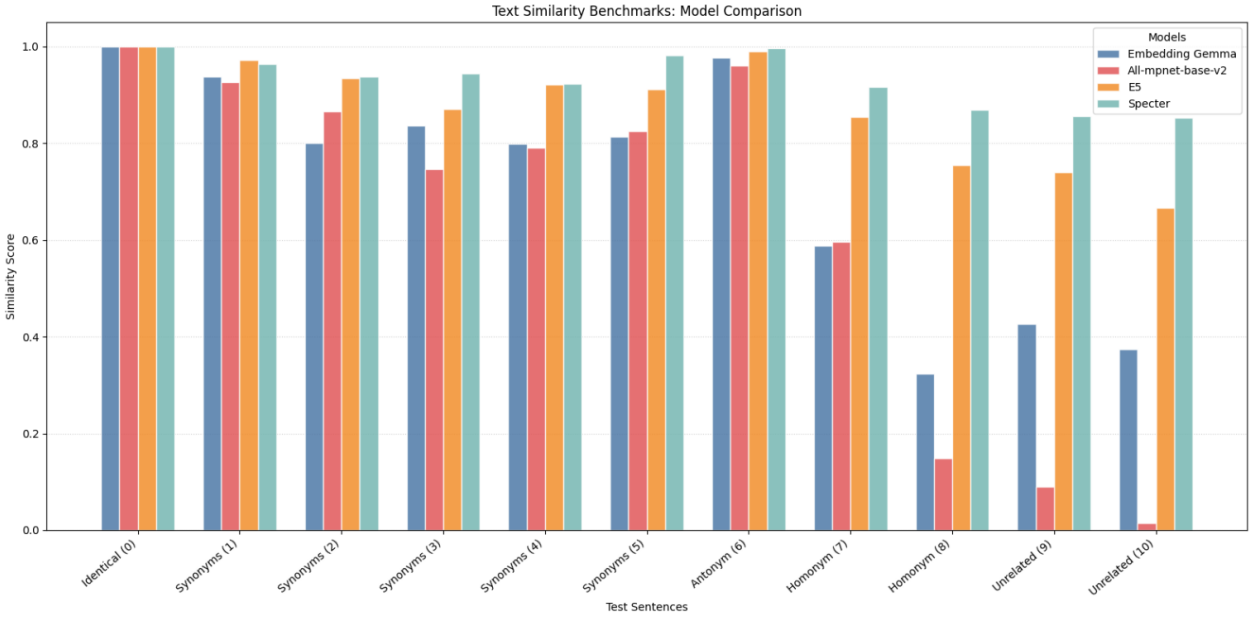


Figure 8: Raw Comparison of Similarity Scores across Embedding Models (SPECTER, E5, all-mpnet-base-v2, and Embedding Gemma).

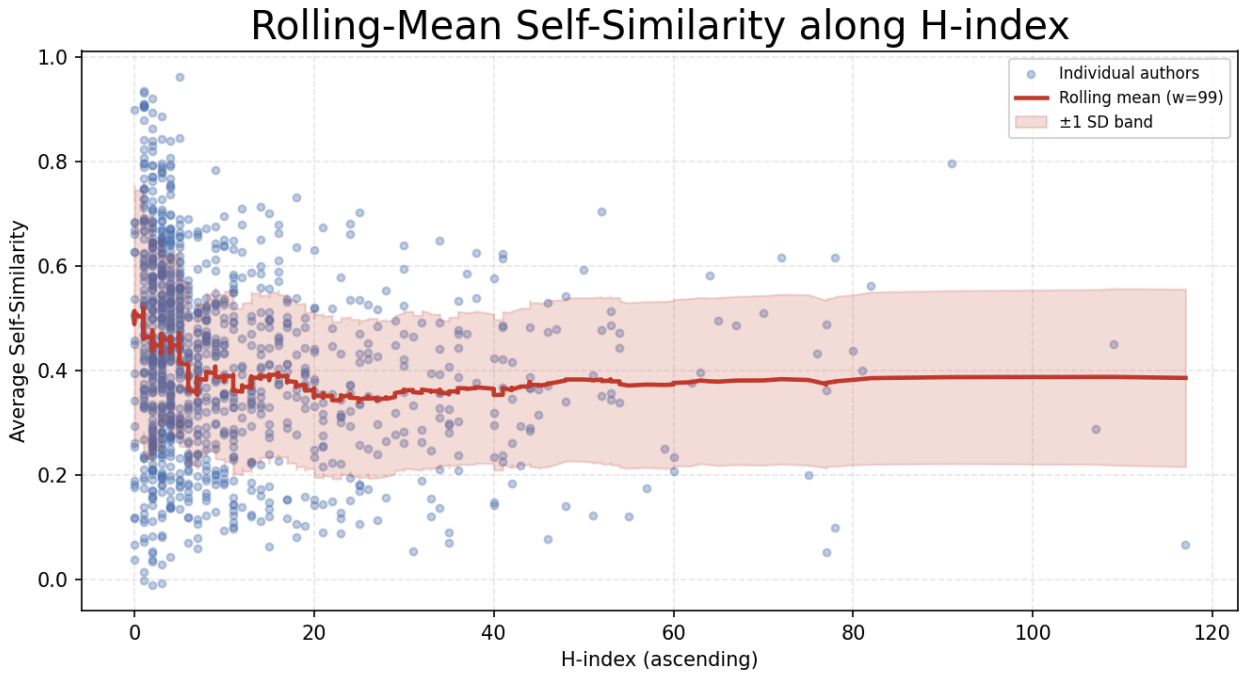


Figure 9: Average intra-author Self-Similarity versus H-index for all included McGill Physics researchers (grey scatter points). The red curve is a rolling-window mean computed over authors sorted by ascending H-index (window size shown in the plot title); the shaded band is ±1 standard deviation around that mean.